



healthiar

**An R package to quantify and monetize health impacts
attributable to (environmental) exposure**

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the European Union

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How can we quantify the health impact of
air pollution in the Swiss population?



Hello, I'm *healthiar*!



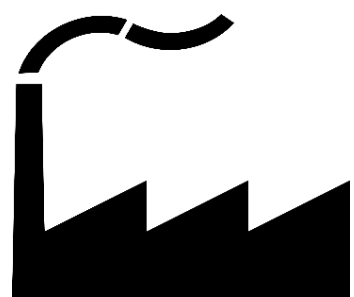
**R package to quantify and monetize
health impacts attributable to
(environmental) exposure**



Example: Lung cancer cases attributable to PM2.5 exposure



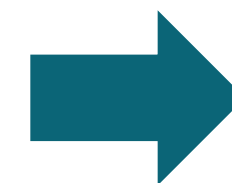
Baseline health
data (BHD)
Incidence =
4800 cases



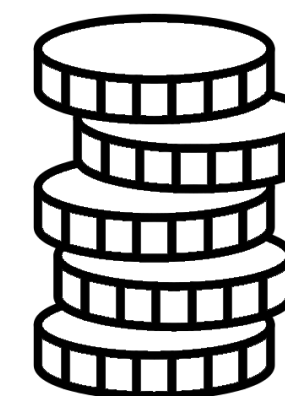
Exposure
= 8.85 $\mu\text{g} / \text{m}^3$



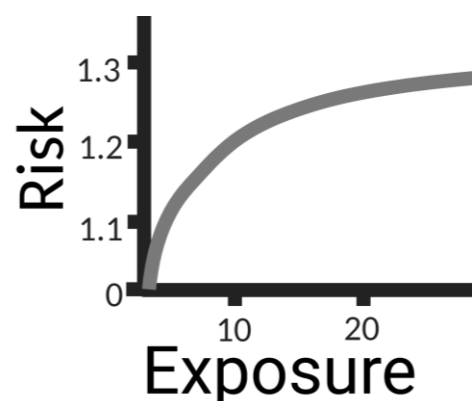
```
attribute_health(  
  bhd_central = 4800,  
  exp_central = 8.85,  
  cutoff_central = 5,  
  rr_central = 1.277,  
  rr_increment = 10,  
  erf_shape = "log_linear")
```



Attributable
health impact
= 431 cases



Direct costs
= 10.8 Mio. CHF



Exposure-
response
function (ERF)
RR = 1.277

Swiss TPH



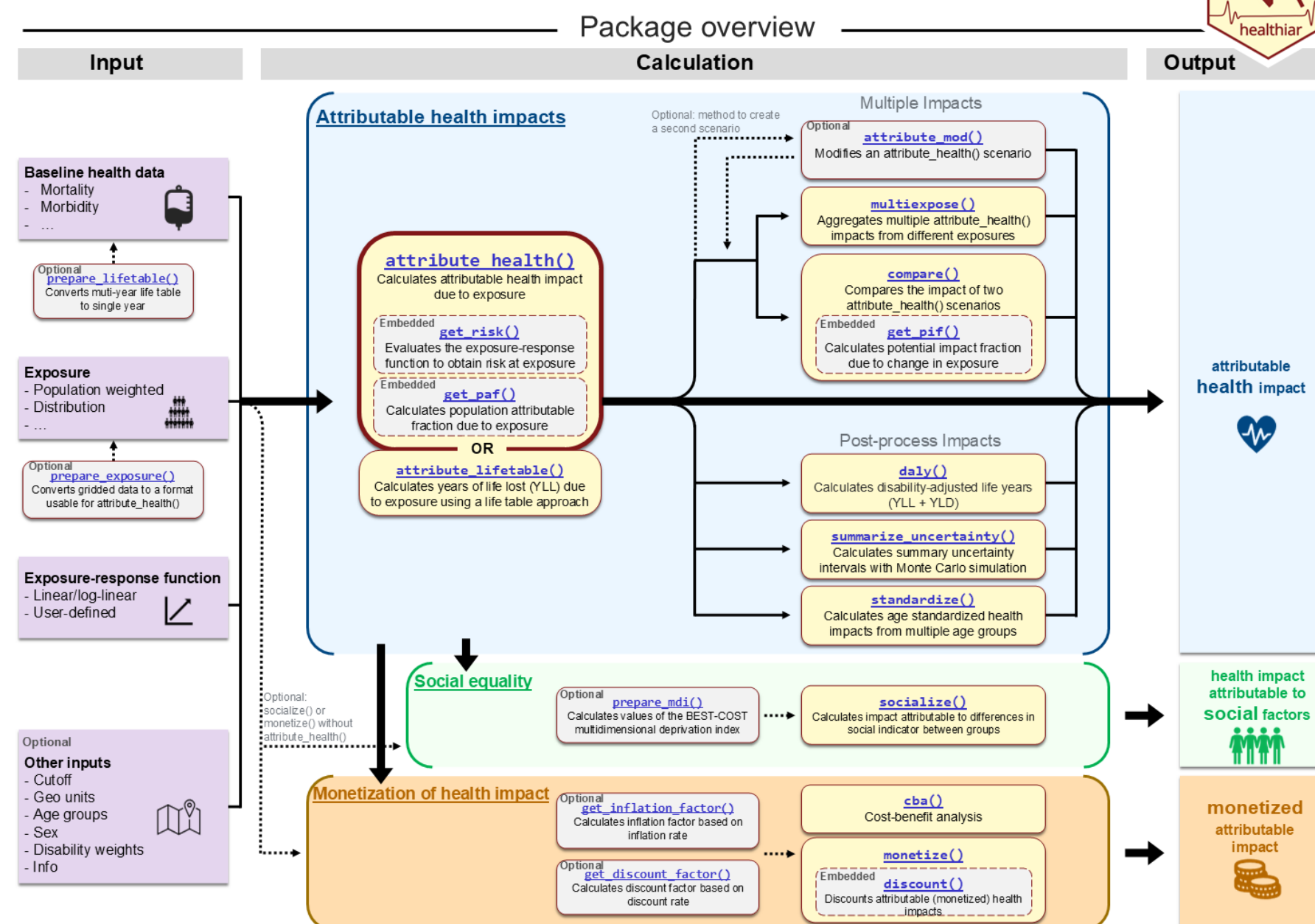


Why use *healthiar*?

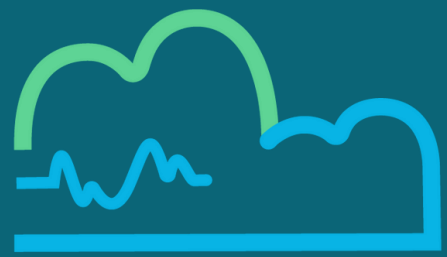
- Sound methodology & documentation
- Open-source & available on CRAN
- >20 functions → many calculation pathways
 - Relative & absolute risk, custom ERF, CIs, subgroup analysis, categorical exposure, life table, ...
- Save time & energy 😊

💡 Among other projects, *healthiar* was used in BEST-COST, UBDPolicy and in an EEA report on noise¹

healthiar : : CHEAT SHEET

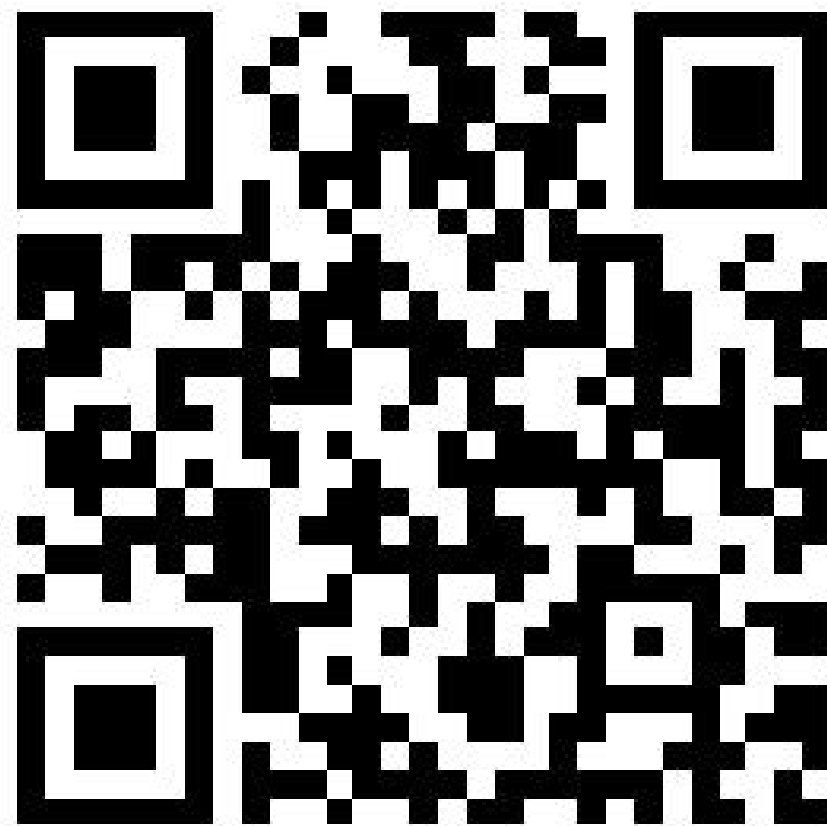


¹ ETC HE Report 2023/11 (Engelmann et al.)



Take-home message

***healthiar* is a ready-to-use R tool to help you quantify attributable health impacts**



healthiar website

Thank you!

Any ideas for new
features? Let us know 😊

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 **BEST-COST**